ISSUE 01 · POLAR / ICE CITY · ECO-POLIS 3D BUILDER

AURORA HAVEN

A polar city that thrives

The city's own sun

A compact fusion station keeps 45,000 people warm through the polar night

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A warm heart inside the ice

Four geothermal spa levels hidden in a 35-metre snow peak

PAGE 12

Bread and water from snow and air

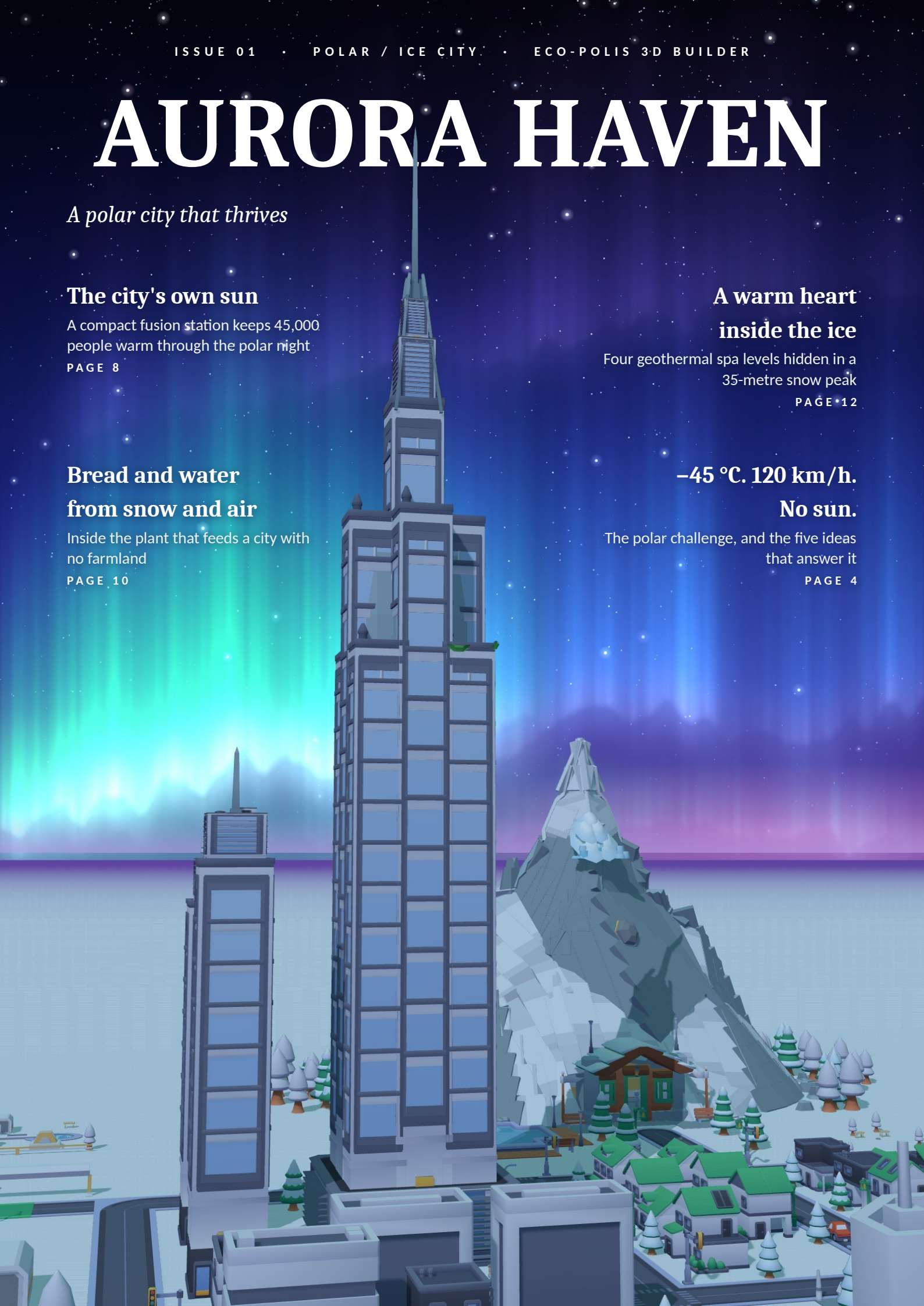
Inside the plant that feeds a city with no farmland

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−45 °C. 120 km/h. No sun.

The polar challenge, and the five ideas that answer it

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Until next time



The Aurora Lagoon and the North Portal of Mount Aurora under the northern lights.

FROM THE EDITORS

Welcome to the first issue of Aurora Haven, a magazine about our city for the Polar / Ice environment.

Aurora Haven is a compact city of about 45,000 people on a frozen tundra plateau, below a 35-metre snow peak called Mount Aurora. Our goal was not only to survive the cold but to build a city where people really want to live.

In these pages you will find the challenge we faced, our design ideas, our three signature districts and everyday life in the city. At the back, a self-guided tour shows you how to explore Aurora Haven yourself in 3D.

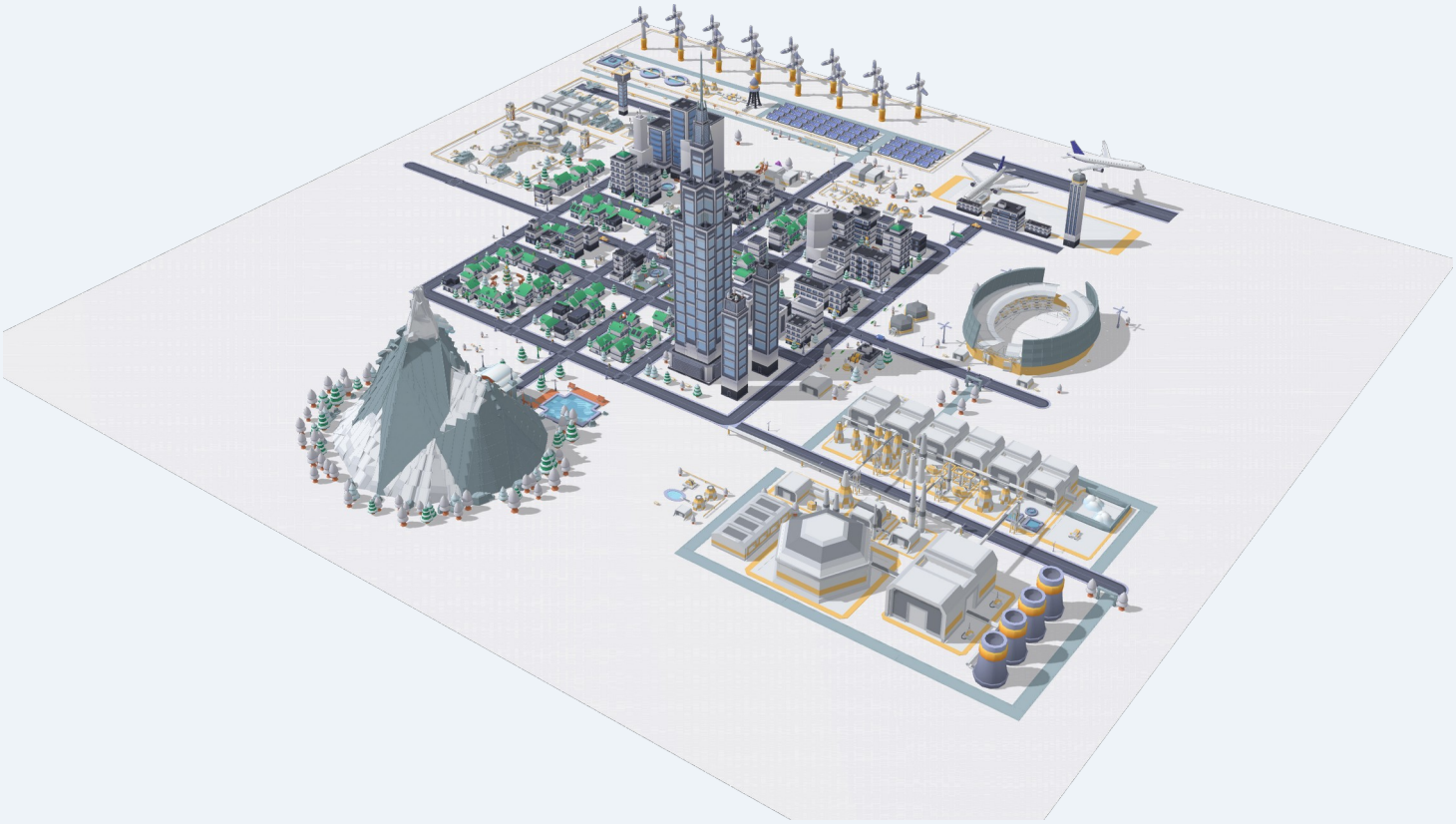
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ON THE COVER

Aurora 81 – Polar Landmark Tower, named in honour of Landmark 81 in Ho Chi Minh City, with Mount Aurora behind it under the northern lights. Its nine bundled glass tubes step back like bamboo stalks and cut blizzard wind load by about 25%.

Welcome to Aurora Haven



Aurora Haven seen from the south-east, rendered from the Eco-Polis 3D city file.

Aurora Haven is a compact polar city of around 45,000 people, built on a frozen tundra plateau beneath the snow peak of Mount Aurora. Winters fall to −45 °C, the sun disappears for two months of polar night, and almost nothing can grow. Aurora Haven is designed not just to survive these conditions but to make life here warm, healthy and sustainable.

The whole city is shaped around one idea: heat, energy, water and food should never be wasted. Every building shares a single heating network, the city runs on zero-carbon energy, and nearly all of its water and waste is recycled.

Homes, schools, healthcare, shops and parks sit within a short walk of each other, connected by a downtown metro, electric trams and heated streets.

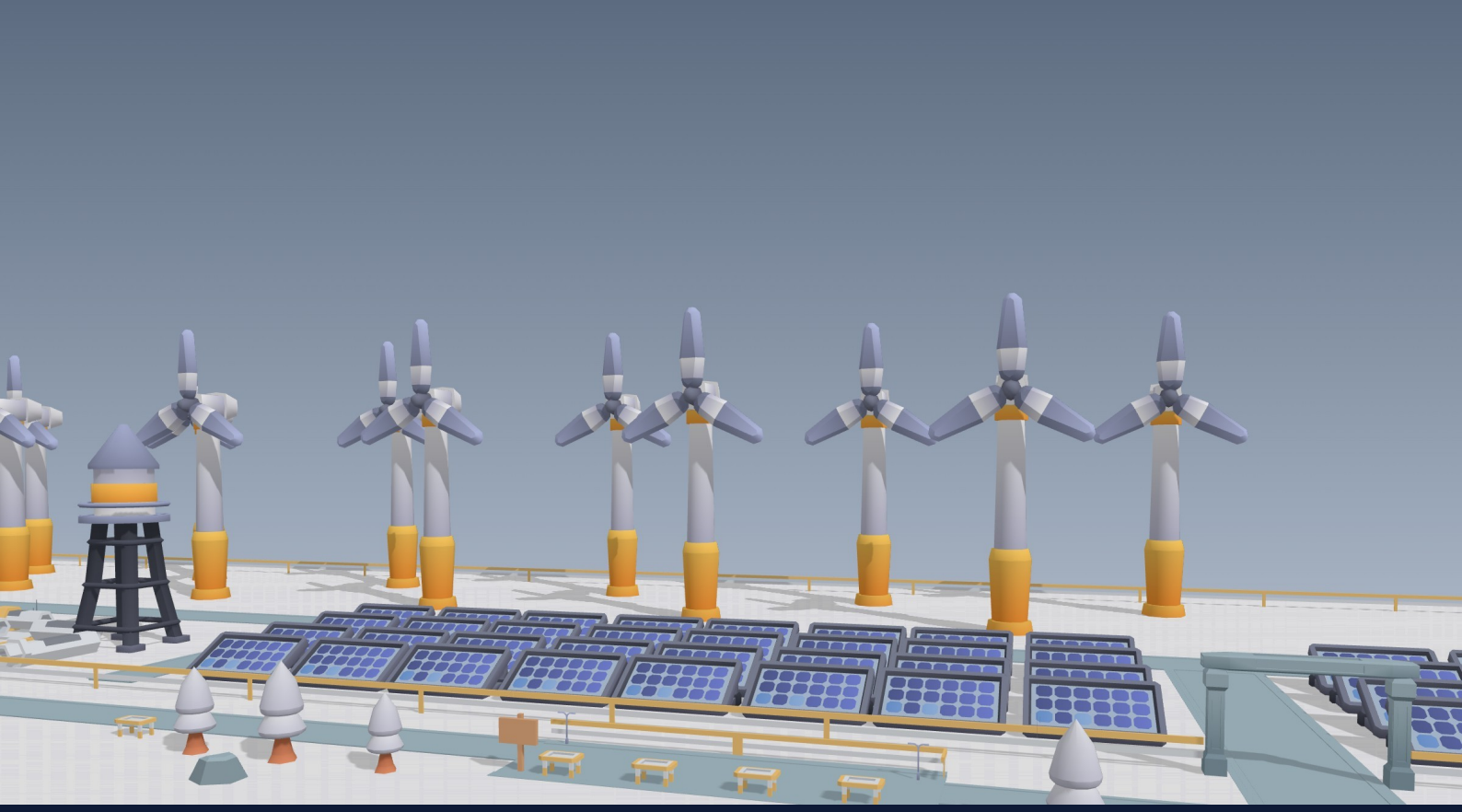
What makes Aurora Haven unique is its three signature districts. A fusion power station acts as the city's own sun, giving steady power and heat even through the polar night. A Snow-to-Food & Water Plant turns snow and air into clean drinking water and food, so the city can feed itself without farmland. And inside Mount Aurora, a geothermal spa resort offers warmth, rest and a view of the northern lights. Together they form one closed loop that keeps the city powered, fed and happy in one of the harshest places on Earth.

45,000
residents

−45 °C
winter lows

35 m
Mount Aurora, the snow peak

10 min
walk to daily needs



The windy northern edge, with the Ice-Proof Wind Farm on the horizon.

THE PLACE

The Polar Challenge

One of the hardest places on Earth to build a city, so every building had to answer one question.

Aurora Haven sits in one of the hardest environments on Earth. In winter the temperature falls to minus 45 degrees. For about two months the sun never rises, and for two months in summer it never sets. Blizzards can reach 120 kilometres per hour.

The ground is permafrost, so a warm building can thaw it and crack its own foundations. There is almost no farmland, and in deep winter supplies can only arrive by air.

So the designers asked one question about every building in the city:

"How does it help people stay warm, fed, safe and happy here?"

−45 °C
winter temperatures

2 months
of polar night, and two of midnight sun

120 km/h
blizzards

Permafrost
frozen ground that heaves under heat

Five Design Principles

Heat, energy, water and food should never be wasted.



Downtown around the Aurora Roundabout, with the Snow-to-Food plant and Mount Aurora behind.

1

Heat is life

One district-heating loop, fed by the fusion station and the geothermal plant, warms every building. Triple-glazed buildings on insulated piles do not thaw the permafrost.

2

Zero carbon

No fossil fuel burns anywhere in the city. Power comes from fusion, wind, sun and geothermal heat.

3

Closed loops

About 95% of water is recycled and rubbish is reused, so water, food and waste go round instead of being thrown away.

4

Compact and car-light

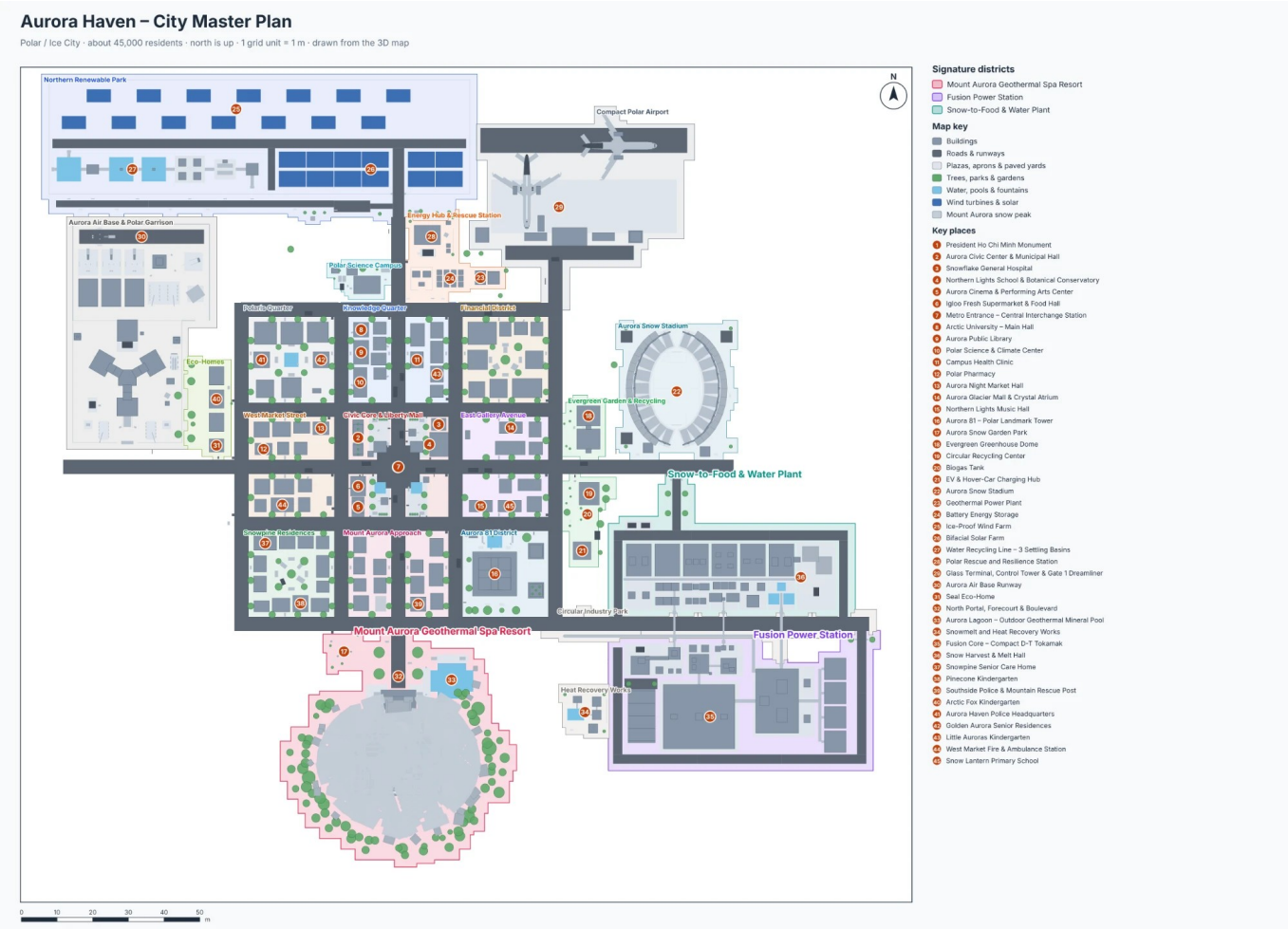
Daily needs are within a 10-minute walk, and the few cars are electric.

5

Resilient and healthy

Shelters keep every resident safe for 30 days, while light, parks, culture and a spa carry people through the polar night.

The Whole City Plan



Aurora Haven – City Master Plan. North is up; one grid unit is one metre. Drawn from the 3D map.

DOWNTOWN

Nine blocks around the Aurora Roundabout and the President Ho Chi Minh Monument, with the hospital, schools, shops and the metro interchange.

NORTH

The windy north holds the Northern Renewable Park, the energy hub, the Polar Rescue and Resilience Station and the airport.

WEST

The air base and garrison, behind a belt of trees that screens the eco-homes.

EAST

Greenhouses, the snow stadium, the fusion station and the Snow-to-Food plant, away from homes but linked by pipe bridges and heat mains.

SOUTH

Framed by Liberty Mall, Mount Aurora rises with its geothermal spa resort inside.

SIGNATURE DISTRICTS

- **Fusion Power Station**
- **Snow-to-Food & Water Plant**
- **Mount Aurora Geothermal Spa Resort**

Nine Blocks, One Short Walk



Looking south down Liberty Mall from the Aurora Roundabout to Mount Aurora.

Downtown is laid out in nine blocks around the Aurora Roundabout. Shops, schools, healthcare and homes are within a short walk of each other, and the metro interchange lies under Liberty Mall. The map below follows the city plan, with north at the top.

Polaris Quarter

Aurora Core – AI Control Tower · Aurora Haven Police Headquarters · Polaris Green Office Tower

Knowledge Quarter

Arctic University – Main Hall · Aurora Public Library

Financial District

Arctic Stock Exchange · Aurora Green Finance Hall · Snowline Courthouse · Aurora Business Hotel

West Market Street

West Market Fire & Ambulance Station · Aurora Night Market Hall · Polar Pharmacy

Civic Core & Liberty Mall

President Ho Chi Minh Monument · Civic Center · Snowflake General Hospital · Northern Lights School · Igloo Fresh · Aurora Cinema · Metro

East Gallery Avenue

Aurora Glacier Mall & Crystal Atrium · Snow Lantern Primary School · Northern Lights Music Hall

Snowpine Residences

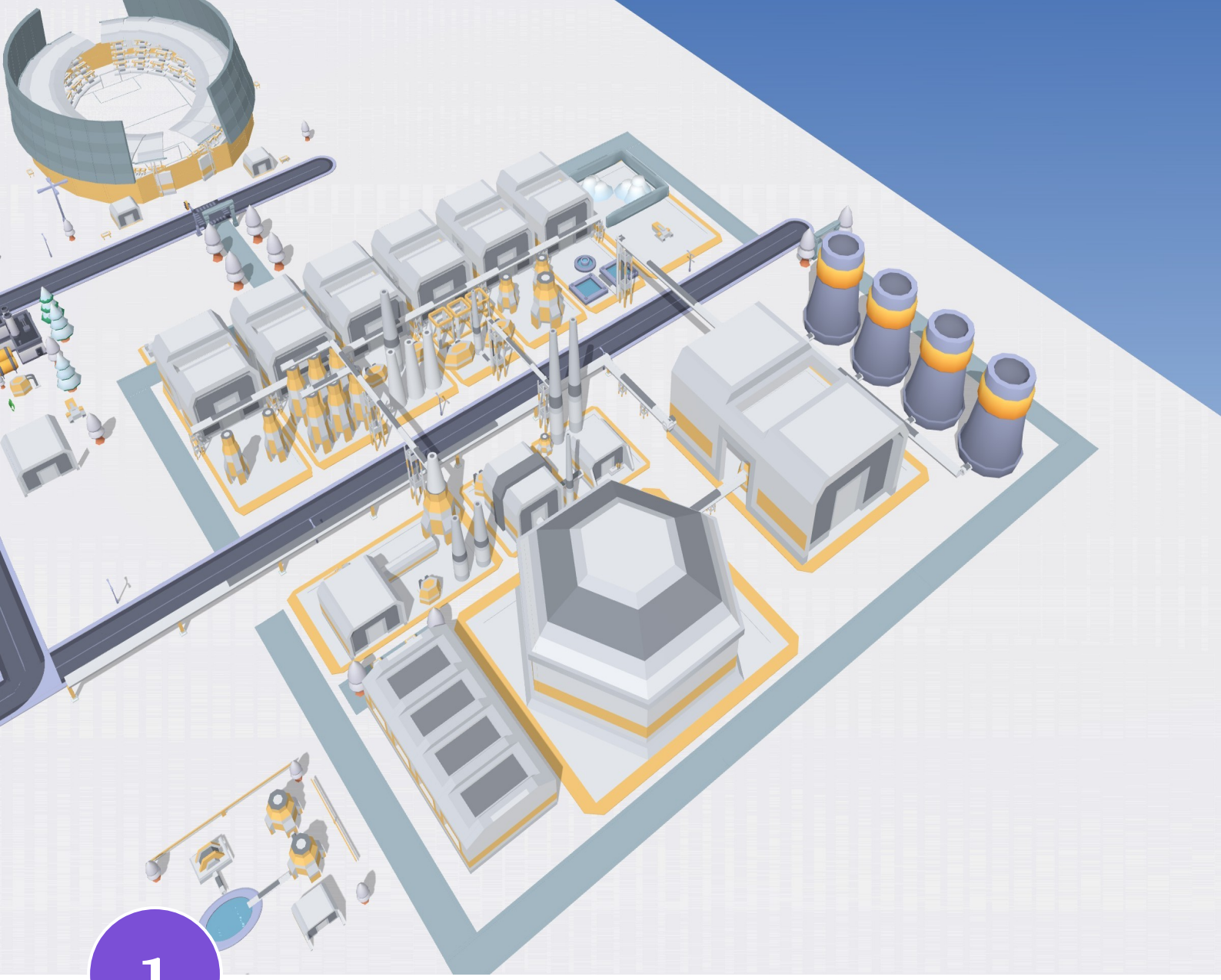
Family homes · Pinecone Kindergarten · Snowpine Senior Care Home

Mount Aurora Approach

Mount Aurora Guesthouse · Southside Police & Mountain Rescue Post · spa shuttle street

Aurora 81 District

Aurora 81 – Polar Landmark Tower · Aurora 81 Residences (Twin Towers)



1

SIGNATURE DISTRICT 1 · FUSION POWER STATION

The City's Own Sun

A compact tokamak gives Aurora Haven about 300 MW of electricity and 450 MW of district heat, steady through the polar night, with no CO_2 and no meltdown.

Our first signature district is the city's own sun. Stage 1 takes deuterium from snowmelt water, so the fuel literally falls from the sky as snow. Stage 2 breeds tritium in a lithium blanket, so none is imported.

Stage 3 is the tokamak, where superconducting magnets hold a 150-million-degree plasma that makes about 800 megawatts of heat.

Stage 4 turns that heat into about 300 megawatts of electricity and sends 450 megawatts into the district-heating loop. Stage 5 turns rubbish into clean gas and building blocks.

The reactor runs on about 170 grams of fuel a day. If anything fails, the plasma simply goes out in a split second, so it cannot melt down or run away.

Five Stages, One Sun



Fusion Power Station – District Plan. Signature district 1: a compact D-T tokamak.

1 **Heavy-water separation**

Takes deuterium from the deuterium-rich water left by the food plant's electrolysers and purifies it to 99.8% heavy water.

2 **Lithium blanket & tritium fuel cycle**

Breeds slightly more tritium than the reactor burns, so none is imported. Recovered helium cools the hospital's MRI scanners.

3 **Fusion core: compact D-T tokamak**

Superconducting magnets hold a 150-million-degree plasma. No CO_2, no long-lived waste, no meltdown.

4 **Power conversion & heat export**

Supercritical-CO_2 turbines make about 300 MW of electricity and send about 450 MW of heat to the city.

5 **Plasma waste converter**

Torches hotter than 5,000 °C turn about 20 t of rubbish a day into clean gas, metals and paving blocks.

C **Control & visitor centre**

Operators and an AI digital twin can stop the plasma in milliseconds; a public gallery faces the reactor.

800 MW
of heat from the plasma

300 MW
of electricity

450 MW
of district heat for the city

170 g
of fuel a day, about 70 g of it deuterium

1 litre
of liquid helium a day for MRI scanners



2

SIGNATURE DISTRICT 2 · SNOW-TO-FOOD & WATER PLANT

Bread and Water from Snow and Air

Six stages turn tundra snow and air into all of the city's drinking water and about half of its protein, with no farmland at all.

What do you eat where nothing grows? Stage 1 melts about 3,000 tonnes of clean tundra snow a day with spare heat from the fusion station, so the snow does the job of a cooling tower. Stage 2 turns the meltwater into all the city's drinking water.

Stage 3 splits water into hydrogen and oxygen and captures CO_2 from the air; Stage 4 turns them into starch and fats. In Stage 5, microbes in six silos make about 2.5 tonnes of protein flour a day. Stage 6 cooks bread, noodles and drinks for the city.

Six Stages from Snow to Supper



Snow-to-Food & Water Plant – District Plan. All drinking water and about half of the city's protein, from snow and air.

1

Snow harvest & melt hall

Electric harvesters gather wind-packed snow upwind of the city; about 14 MW of spare fusion heat melts it.

2

Drinking-water & mineral works

Membranes and UV light remove every germ, calcite adds calcium and magnesium, and sensors test the water every five minutes.

3

Electrolysis, air capture & green ammonia

About 5 t of hydrogen a day; the oxygen goes to the hospital and the waste converter.

4

Carbon-fixation chemistry

A copper catalyst makes methanol, and enzymes build it into starch about eight times faster than a maize field.

5

Bioreactor silos

Hydrogen-eating bacteria make protein flour; yeasts and algae make oils and vitamins, 24 hours a day.

6

Food kitchen, bottling & distribution

Bread, noodles, protein mince and dairy-style drinks, delivered overnight in heated electric vans.

3,000 t

of snow melted a day

2,500 m³

of drinking water a day, all of the city's tap water

20 t

of CO_2 captured from the air a day

2.5 t

of protein flour a day, as much as 600 hectares of soybeans

30 days

of meals and water for everyone in the Food Reserve Bunker



3

SIGNATURE DISTRICT 3 · MOUNT AURORA GEOTHERMAL SPA RESORT

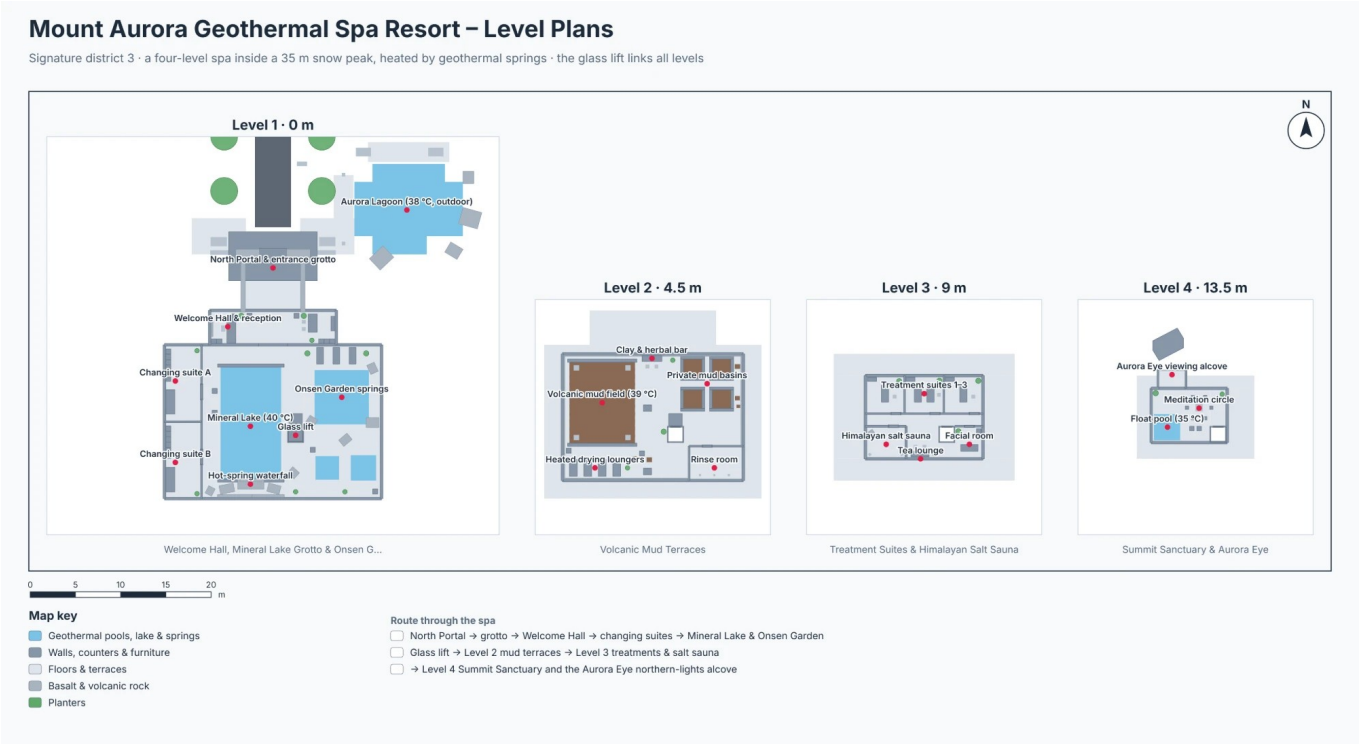
A Warm Heart Inside the Ice

Inside a 35-metre snow peak shaped like Nepal's Ama Dablam, four geothermal spa levels stay at 28 °C all winter.

Guests enter through the North Portal, where a thermal revolving door and an air curtain keep the −45 °C air out. The peak's deep snow works as a blanket that holds in the geothermal heat, and sensors release the snow load safely before an avalanche can build.

Outside is the Aurora Lagoon, a 38-degree mineral pool for bathing under the northern lights. The spa fights the winter blues, brings tourists and income, and in an emergency can shelter up to 1,000 people.

Four Floors of Warmth



Mount Aurora Geothermal Spa Resort – Level Plans. The glass lift links all four levels.

LEVEL 1 · 0 M

Welcome Hall, Mineral Lake Grotto & Onsen Garden

A 40 °C silica-rich spring rises into a 7 × 13 m Mineral Lake; hot, warm and glacier-cold pools boost circulation.

LEVEL 2 · 4.5 M

Volcanic Mud Terraces

An 8 × 10 m mud field held at 39 °C. The mud is filtered and UV-sterilised every night and reused for months.

LEVEL 3 · 9 M

Treatment Suites & Himalayan Salt Sauna

Hot-stone massage, arctic kelp wraps and a 75 °C pink salt sauna heated by the geothermal loop.

LEVEL 4 · 13.5 M

Summit Sanctuary & Aurora Eye

A 35 °C float pool, a meditation circle and a heated glass bubble for watching the northern lights.

28 °C inside the spa

40 °C Mineral Lake spring

39 °C volcanic mud

75 °C salt sauna

35 °C float pool

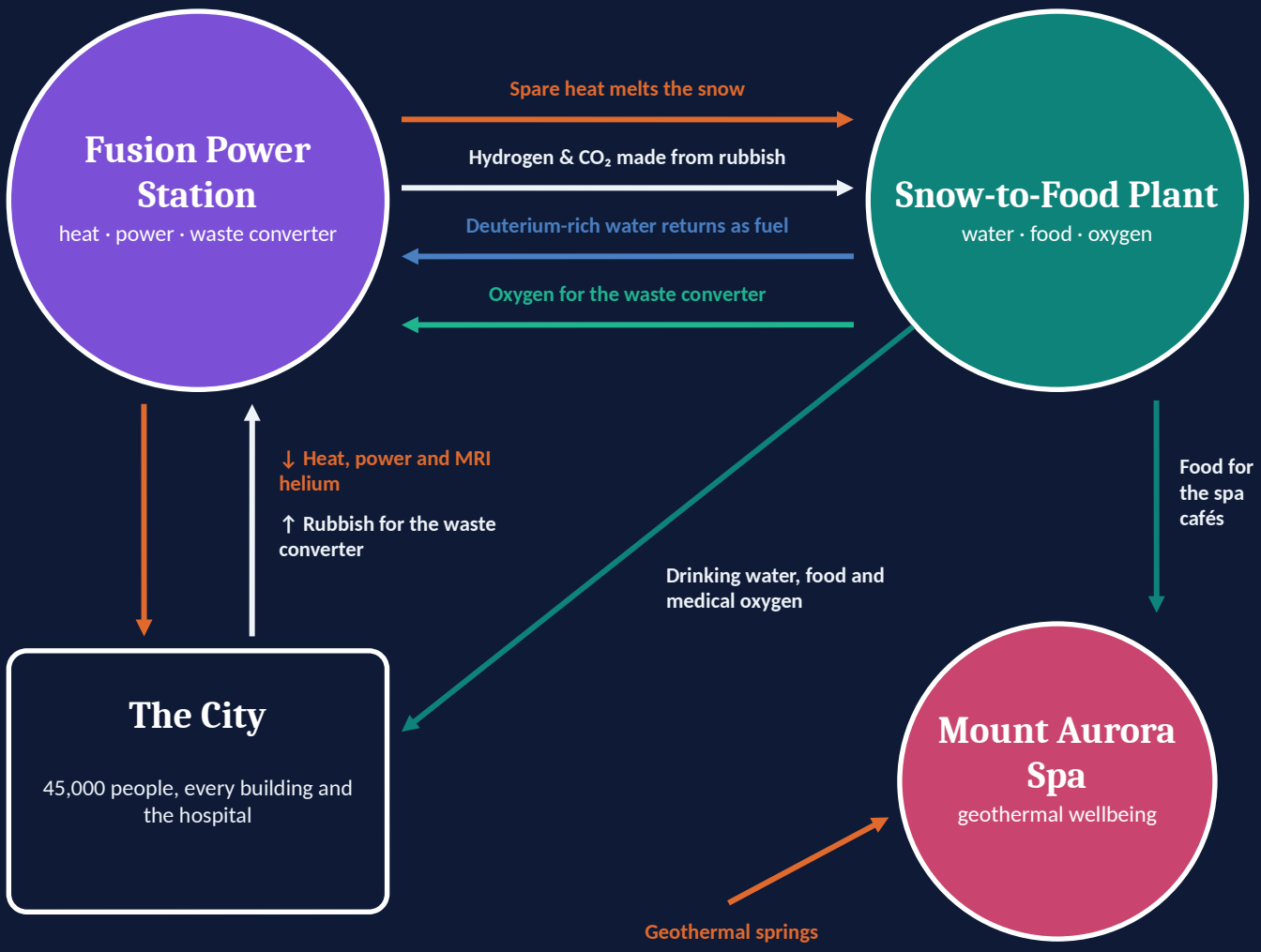
38 °C Aurora Lagoon

Route through the spa: North Portal → grotto → Welcome Hall → changing suites → Mineral Lake & Onsen Garden → glass lift → mud terraces → treatments & salt sauna → Summit Sanctuary and the Aurora Eye.

THE BIG IDEA

Three Districts, One Loop

The three signature districts do not work alone. Heat, water, oxygen and waste go round from one to the next instead of being thrown away.



Heat

Spare heat from the fusion station melts the snow at the food plant, so the snow does the job of a cooling tower.

Water

The food plant's electrolysers leave slightly deuterium-rich water, which goes back to the fusion station as fuel.

Oxygen

The same electrolysers supply oxygen to the plasma waste converter and the hospital.

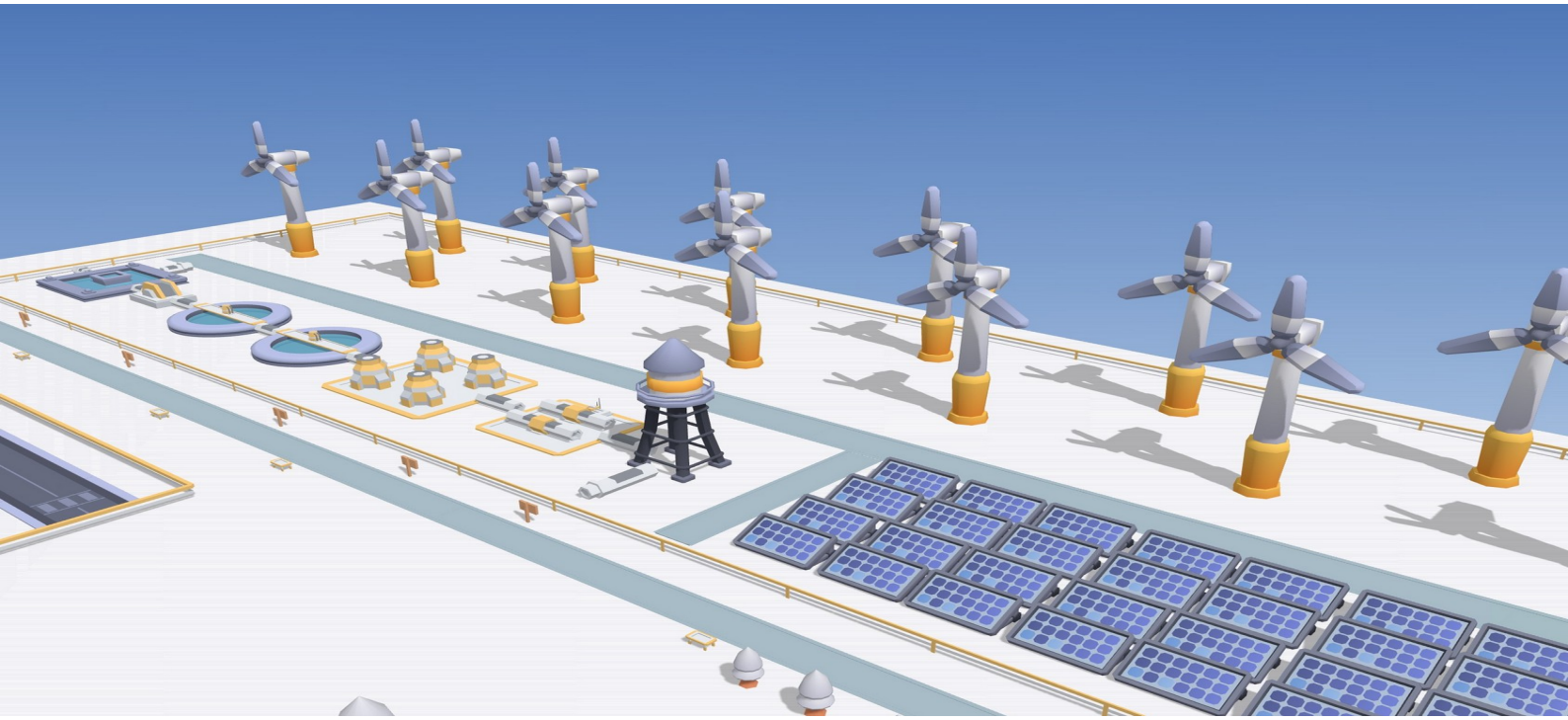
Waste

The converter turns rubbish into hydrogen and CO_2 for the food lines, and its slag becomes paving.

Wellbeing

The spa turns geothermal heat into rest and health, and its restaurants serve food from the plant.

Energy for the Polar Night



Northern Renewable Park: the Ice-Proof Wind Farm and the Bifacial Solar Farm.

The sun disappears for two months, so the city uses sources that cover for each other.

Steady

Fusion

Steady power and heat all year, even in total darkness.

60 MW

Ice-proof wind farm

Heated blades keep turning in freezing fog; most power comes in stormy winter.

+25%

Bifacial solar farm

Snow reflects light onto the panel backs; 24-hour power in the midnight-sun months.

24/7

Geothermal plant

Clean electricity and hot water around the clock, and the second heat source.

200 MWh

Battery bank

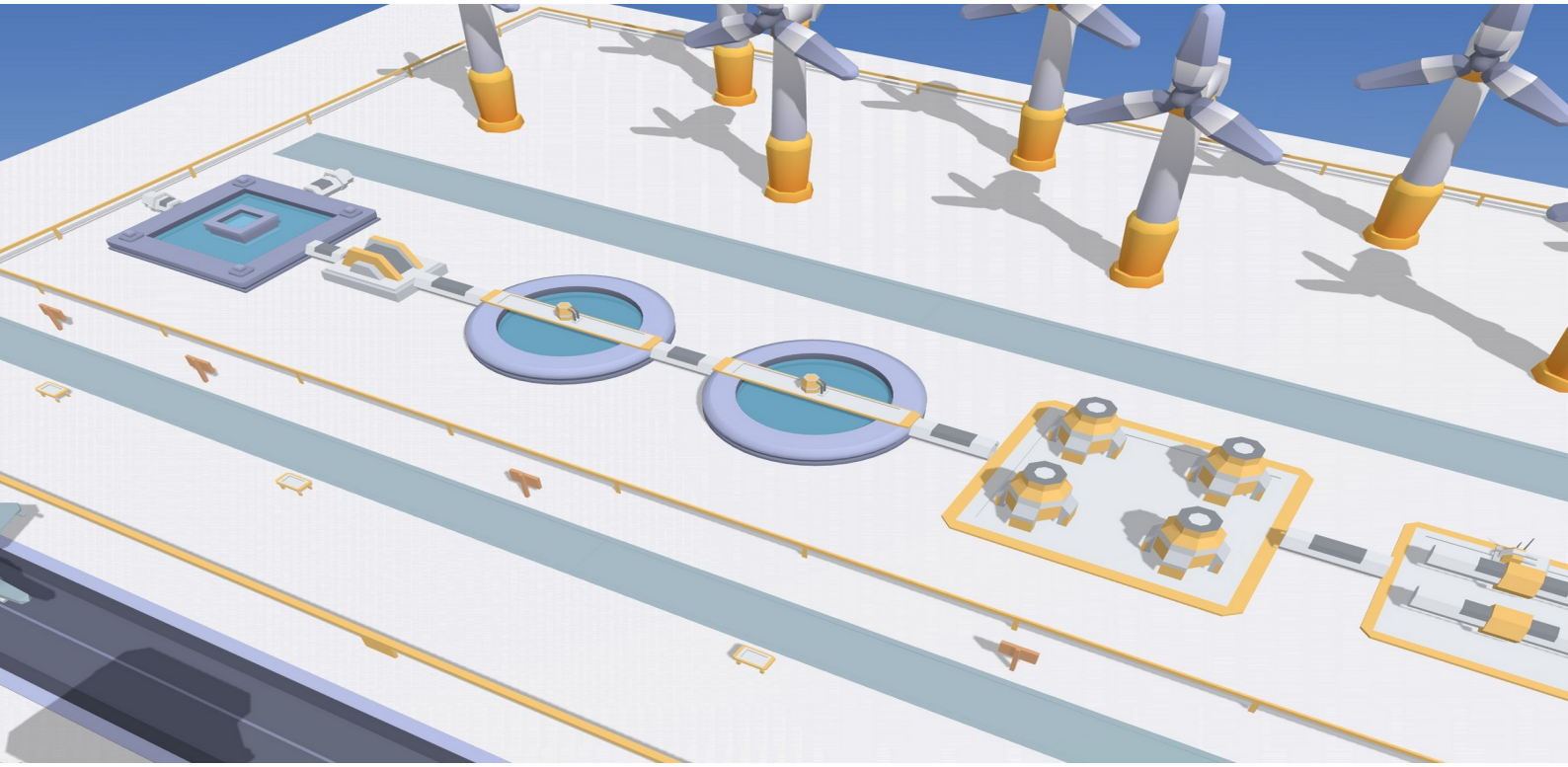
Recycled car batteries store spare wind and solar power for calm, dark days.

12 h

Heat storage tank

Keeps homes warm if a heat source stops, losing less than 1% of its heat a day.

Nothing Is Wasted



The Water Recycling Line in the Northern Renewable Park, with the wind farm behind.

THE SIX-STAGE WATER RECYCLING LINE

1. **Snowmelt & greywater intake**
2. **Intake pumps & bar screen**
3. **Settling basins**
4. **Sand & membrane filters**
5. **UV disinfection**
6. **Recycled water tower**

In Aurora Haven, water goes round and round. Clean snow becomes drinking water at the Snow-to-Food plant. Used greywater from homes flows to the Water Recycling Line: screens, settling basins, sand and membrane filters, UV light, and finally a tower that feeds toilets, laundries, greenhouses and fire hydrants.

Dirty street snow is melted separately and never reaches the taps.

The Circular Recycling Center sorts about 30 tonnes of materials a day with AI optical sorters and robot arms. Food scraps become biogas, and the Plasma Waste Converter handles what is left, turning it into clean gas and building blocks.

Nothing goes to landfill in the permafrost.

95%
of water recycled

30 t
of materials sorted a day

20 t
of rubbish a day turned into gas and blocks

0
landfill in the permafrost

Getting Around Fossil-Free

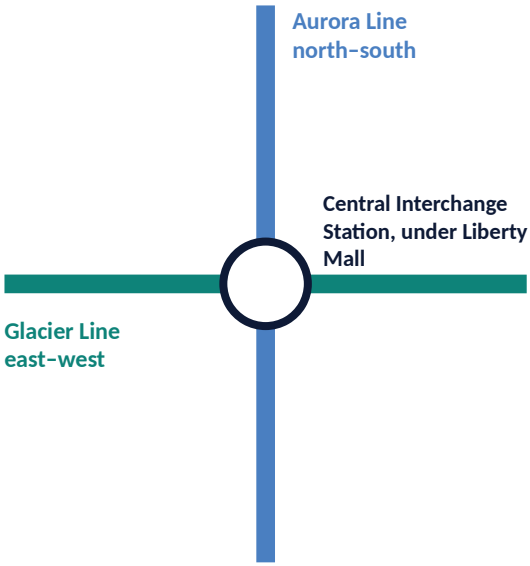


Compact Polar Airport: the glass terminal, the control tower and Flight VN-81 taking off.

Transport in Aurora Haven is electric and built for the cold. The heart of it is the metro: the north–south Aurora Line and the deep east–west Glacier Line meet at the Central Interchange Station under Liberty Mall. The driverless trains run underground, so blizzards never stop them.

On the streets there are electric trams, buses and shared robotaxis, and people wait in heated bus shelters. The roads are warmed from below, so no salt is needed, and an outer ring keeps trucks out of downtown.

Electric cars charge in the heated bays of the EV & Hover-Car Charging Hub, and the airport, with its heated runway and e-fuel planes, keeps the city supplied through deep winter.



3 min
Aurora Line trains at rush hour

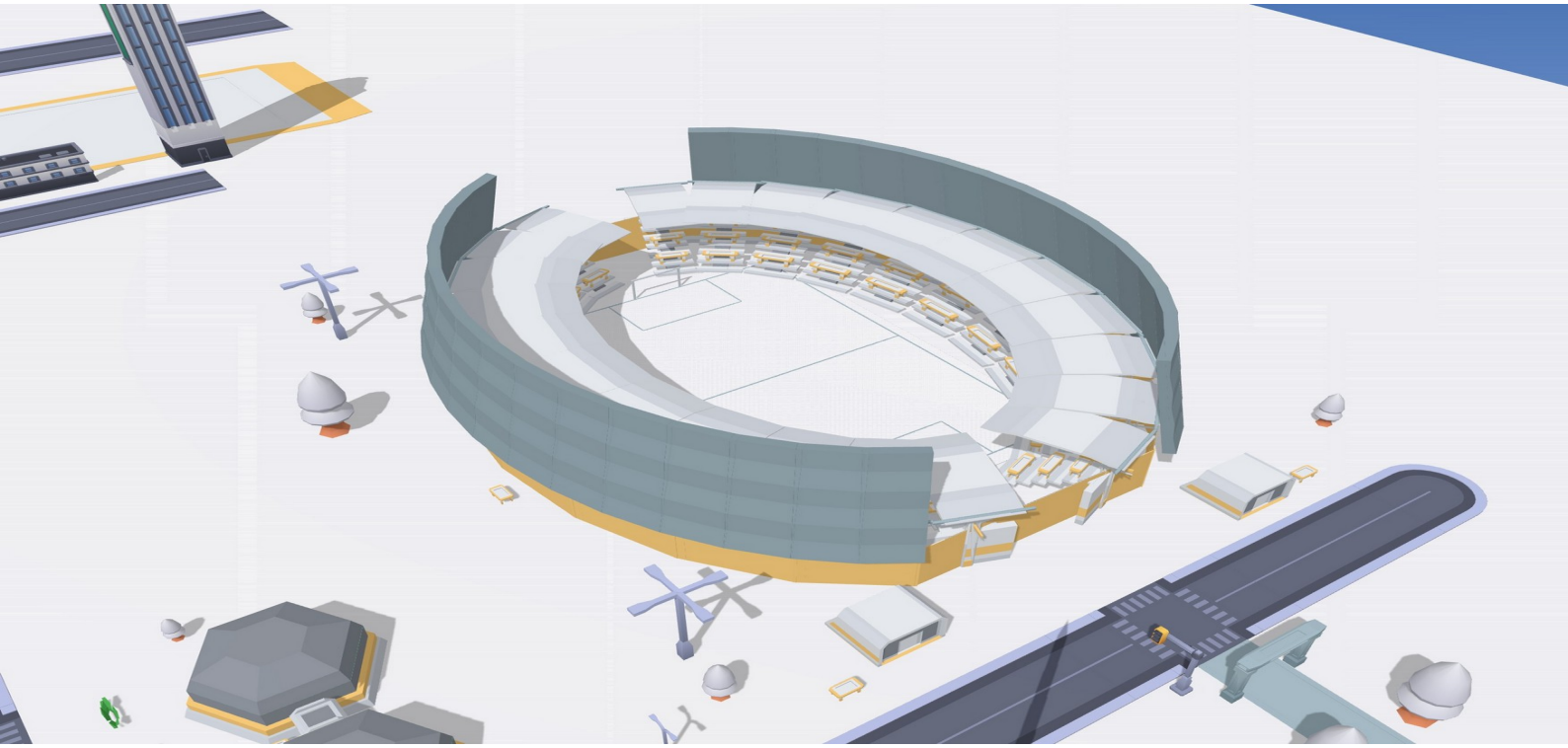
4 min
Glacier Line; every 90 s after big stadium events

30%
of train energy returned by braking

2,800 m
heated airport runway

20 km/h
downtown speed limit

A Livable City Every Day



Aurora Snow Stadium, home of school sports days and the Midwinter Aurora Festival.

HOMES

Family houses in Snowpine Residences, the western eco-homes, towers such as the Aurora 81 Residences and Ice Crystal Apartments, co-living, student and senior housing.

HEALTH

Snowflake General Hospital with about 250 beds and a 24-hour emergency department, a campus clinic and a 24-hour pharmacy.

CULTURE AND PARKS

Aurora Cinema, a music hall, a library open 24 hours in the polar night, the Snow Stadium and parks with heated benches.

LEARNING

Three kindergartens, Snow Lantern Primary School, the Northern Lights School for ages 6 to 18, and Arctic University.

SHOPPING

Igloo Fresh Supermarket, Tundra Grocery, the night and fish markets and the Aurora Glacier Mall with its indoor atrium park.

250
hospital beds

900
pupils at Northern Lights School

5,000
students at Arctic University

8,000
seats in the Snow Stadium

Safe and Ready for Any Storm



The Polar Rescue and Resilience Station in the Energy Hub, north of downtown.

In the polar night, safety comes first. The Polar Rescue and Resilience Station is home to the fire brigade, ambulances, ice rescue and the 112 emergency centre, and its vehicles leave within 60 seconds.

The West Market Fire & Ambulance Station reaches any downtown address in about four minutes, and a rescue helicopter is ready day and night.

Under Liberty Mall, four underground shelters can keep every resident safe for 30 days. Aurora Core, the AI control tower, forecasts blizzards up to 48 hours ahead and alerts every phone.

Cafés and lobbies act as warm points, and the air base runway is a backup for supply flights.

60 s
for rescue vehicles to leave

4 min
to any downtown address

10 min
for the rescue helicopter to be airborne

4
underground shelters

30 days
of supplies for every resident

48 h
blizzard forecasts

EXPLORE

Take the Tour: Eight Stops in 3D

Open the city file in Eco-Polis 3D Builder and click Full Screen. The app locks into View mode, so nothing can move by accident. Then click each place to open its card.

1. **President Ho Chi Minh Monument**
   The civic heart of the city on the Aurora Roundabout.
2. **Metro Entrance – Central Interchange Station**
   Escalators down to the metro and the shelters.
3. **Snowflake General Hospital**
   About 250 beds and a 24-hour emergency department.
4. **[Fusion Power Station – Stage 3] Fusion Core – Compact D-T Tokamak**
   The city's own sun.
5. **[Snow-to-Food & Water Plant – Stage 1] Snow Harvest & Melt Hall**
   Melts about 3,000 t of snow a day with spare fusion heat.
6. **[Mount Aurora] Ama Dablam-Style Snow Peak & Hanging Glacier**
   The 35 m peak with the spa inside.
7. **[Northern Renewable Park] Ice-Proof Wind Farm**
   Two rows of turbines making about 60 MW.
8. **Polar Rescue and Resilience Station**
   Home of the 112 emergency centre.

If there is time: the Aurora Lagoon, the Bifacial Solar Farm and the Recycled Water Tower.

BEHIND THE BUILD

3,700
3D items arranged to one scale

241
named places, each with a Name, Purpose and Special Features card

1 m
per grid unit on every plan

MEETING THE BRIEF

1. **Smart design**
   polar answers to cold, darkness, wind and permafrost
2. **Sustainability**
   clean energy, closed loops, green spaces, electric transport
3. **Livability**
   homes, schools, hospital, shops, parks, cinema, roads
4. **Detail**
   241 places annotated
5. **Timeline**
   built in rounds; final .json submitted 17 October 2026

Places to Visit

#	Name of the place (as in the app)	District	Purpose
1	**President Ho Chi Minh Monument**	Civic Core & Liberty Mall	The city's civic heart on Aurora Roundabout: ceremonies, national holidays and walking tours.
2	**Aurora Civic Center & Municipal Hall**	Civic Core & Liberty Mall	Town hall for the city council, ID cards, permits, housing and family services.
3	**Snowflake General Hospital**	Civic Core & Liberty Mall	Main hospital with about 250 beds and a 24-hour emergency department.
4	**Northern Lights School & Botanical Conservatory**	Civic Core & Liberty Mall	Primary and secondary school for about 900 pupils aged 6–18.
5	**Igloo Fresh Supermarket & Food Hall**	Civic Core & Liberty Mall	Downtown's main supermarket and food court; about half its food is made in the city.
6	**Aurora Cinema & Performing Arts Center**	Civic Core & Liberty Mall	Cinema and theatre for films, concerts, ballet and school shows.
7	**Underground Metro & Emergency Shelters**	Civic Core & Liberty Mall (underground)	Metro interchange linked to four shelters that last 30 days.
8	**Aurora Core – AI Control Tower**	Polaris Quarter	Operations centre where an AI digital twin runs traffic, heat, power and alerts.
9	**Aurora Haven Police Headquarters**	Polaris Quarter	Central police station with a 24-hour police control room.
10	**Arctic University – Main Hall**	Knowledge Quarter	Lecture halls for about 5,000 students of polar engineering, fusion, food science, medicine and climate.
11	**Aurora Public Library**	Knowledge Quarter	Free library, open 24 hours in the polar night; lends skis, telescopes and light-therapy lamps.
12	**West Market Fire & Ambulance Station**	West Market Street	Downtown fire and ambulance branch that reaches any downtown address in about 4 minutes.
13	**Aurora Glacier Mall & Crystal Atrium**	East Gallery Avenue	Largest downtown mall, with an indoor atrium park of real trees.
14	**Snow Lantern Primary School**	East Gallery Avenue	Primary school for about 450 children aged 6–11 from the southern districts.
15	**Pinecone Kindergarten**	Snowpine Residences	Kindergarten and day nursery for about 40 children aged 1–6.
16	**Aurora 81 – Polar Landmark Tower**	Aurora 81 District	Tallest tower and skyline symbol: offices, hotel and sky apartments, net-zero energy.
17	**Aurora Snow Garden Park**	Foot of Mount Aurora	Public park with heated benches, a sledging slope and a snow-sculpture trail.
18	**[Mount Aurora] Ama Dablam-Style Snow Peak & Hanging Glacier**	Mount Aurora Geothermal Spa Resort	35 m landmark peak with the four-level spa inside.

Places to Visit, Continued

#	Name of the place (as in the app)	District	Purpose
19	**[Mount Aurora – Level 4] Summit Sanctuary & Aurora Eye**	Mount Aurora Geothermal Spa Resort	Float pool, meditation circle and the Aurora Eye viewing bubble for the northern lights.
20	**[Mount Aurora] Aurora Lagoon – Outdoor Geothermal Mineral Pool**	Mount Aurora Geothermal Spa Resort	38 °C open-air silica pool for bathing under the northern lights.
21	**[Fusion Power Station – Stage 3] Fusion Core – Compact D-T Tokamak**	Fusion Power Station	150-million-degree plasma giving about 800 MW of heat; it cannot melt down.
22	**[Fusion Power Station – Stage 4] Power Conversion & Heat Export Hall**	Fusion Power Station	Makes about 300 MW of electricity and sends about 450 MW of heat to the city.
23	**[Fusion Power Station – Stage 5] Plasma Waste Converter**	Fusion Power Station	Turns about 20 t of rubbish a day into clean gas, recovered metals and building blocks.
24	**[Snow-to-Food & Water Plant – Stage 1] Snow Harvest & Melt Hall**	Snow-to-Food & Water Plant	Melts about 3,000 t of clean snow a day with spare fusion heat.
25	**[Snow-to-Food & Water Plant – Stage 2] Drinking-Water & Mineral Works**	Snow-to-Food & Water Plant	Makes about 2,500 m^3 of drinking water a day, all of the city's tap water.
26	**[Snow-to-Food & Water Plant – Stage 5] Bioreactor Silos**	Snow-to-Food & Water Plant	Six silos where microbes make about 2.5 t of protein flour a day.
27	**[Northern Renewable Park] Ice-Proof Wind Farm**	Northern Renewable Park	Two rows of ice-proof turbines making about 60 MW.
28	**[Northern Renewable Park] Bifacial Solar Farm**	Northern Renewable Park	Double-sided panels that make power 24 hours a day in the midnight-sun months.
29	**[Northern Renewable Park] Water Recycling Line – 6 Recycled Water Tower**	Northern Renewable Park	Stores recycled water for toilets, laundries, street cleaning, irrigation and hydrants.
30	**Geothermal Power Plant**	Energy Hub & Rescue Station	Clean electricity and hot water around the clock; the second heat source.
31	**Polar Rescue and Resilience Station**	Energy Hub & Rescue Station	Fire, ambulance and ice-rescue headquarters, home of the 112 emergency centre.
32	**[Compact Polar Airport] Glass Terminal, Control Tower & Gate 1 Dreamliner**	Compact Polar Airport	Passenger and cargo airport; the city's supply link in deep winter.
33	**Evergreen Greenhouse Dome**	Evergreen Garden & Greenhouses	Grows fresh vegetables and berries all year under LED light.
34	**Circular Recycling Center**	Evergreen Garden & Greenhouses	Sorts, repairs and recycles about 30 t of materials a day.
35	**Aurora Snow Stadium**	Aurora Snow Stadium	About 8,000 seats for snow sports, school sports days and the Midwinter Aurora Festival.

KEY FIGURES

Aurora Haven by the Numbers

THE CITY

45,000 residents

241 named places

3,700 3D items

CLIMATE

−45 °C winter lows

2 months of polar night

120 km/h blizzards

FUSION POWER STATION

300 MW electricity

450 MW district heat

99.8% heavy-water purity

SNOW-TO-FOOD & WATER PLANT

5 t hydrogen a day

4 t starch a day

1 t fat a day

ENERGY AND STORAGE

60 MW wind farm

200 MWh battery bank

<1% heat lost a day by the heat tank

MOUNT AURORA SPA

4 levels, 0 to 13.5 m

28 °C all winter

1,000 people sheltered in an emergency

LIVABILITY

600 Aurora 81 apartments

450 pupils at Snow Lantern Primary

3 kindergartens

TRANSPORT

2 metro lines

5 min spa shuttle

20 min airport shuttle

All figures are design values for the Aurora Haven city project.

UNTIL NEXT TIME

Warm, Green and Ready

Aurora Haven does not just survive the polar environment; it thrives in it. Heat from fusion and geothermal energy keeps every building warm, and clean energy and closed loops mean no fossil fuel and almost no waste. Most importantly, it is a place where 45,000 people can live, learn, work and watch the northern lights together.

AURORA HAVEN

[Group name] · [Member names]
[Class] · [Date]
Built in Eco-Polis 3D Builder
Polar / Ice City

